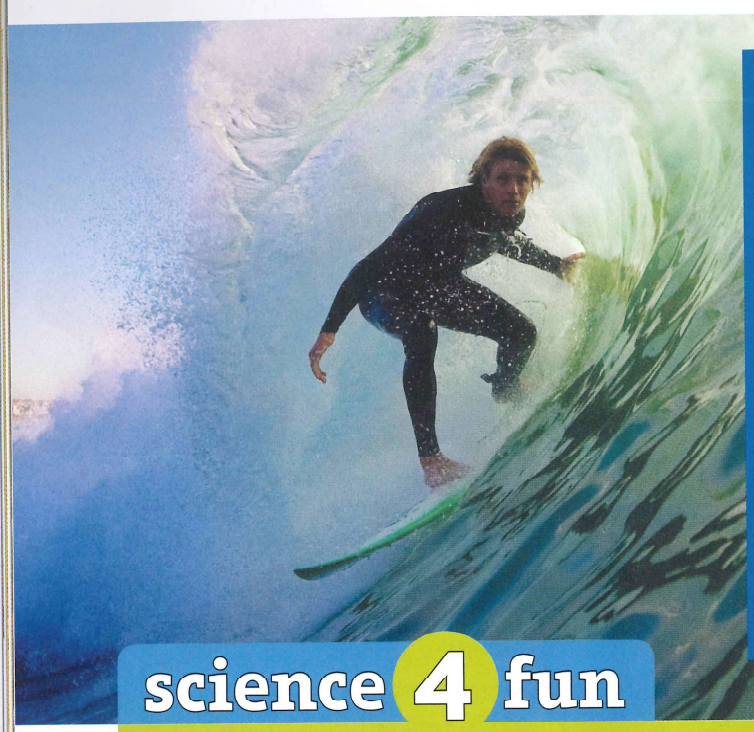
MODULE 7.1

What are forces?



Forces act on you all the time. When you surf, you can feel many forces pushing and pulling you in different directions. Gravity always pulls you down, towards planet Earth. This force can be balanced by the support of the surfboard acting upwards. Forces of the waves push you towards shore, and friction from the air and water pull you back out to sea. Surfing requires your body to balance all those forces acting on you. If they're not balanced, then down you go!

science 4 fun

Unbalanced diet

Can you balance a soft-drink can on its side?

Collect this ...

- empty soft-drink can
- water

Do this ...

This activity is best done outside or on a bench that can easily be wiped down.

1. One-third fill an empty soft-drink can with water.
2. Try to balance the can by placing the ridge at its base on a bench at an angle of about 45°.
3. See if you can let the can go without it tipping over!

Record this ...

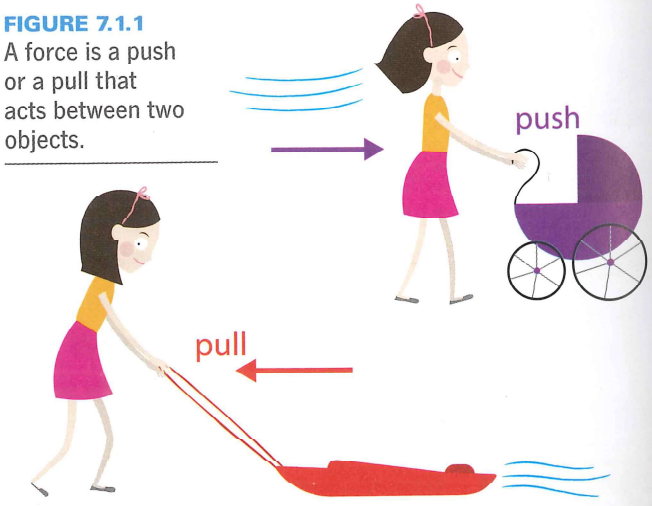
1. Describe what happened to the can.
2. Explain why you think this happened.

Push or pull

A **force** is an interaction between two objects. This force, or interaction can be a push or a pull. Forces always happen in pairs. The girl in Figure 7.1.1 pushes on the handle of the pram. As she does this, the handle of the pram pushes back on the girl's hand. Similarly, when the girl pulls the cord attached to the toy boat, this cord pulls back on the girl's hand.

Force is measured using a unit called the **newton** (symbol N). This unit is named after the English scientist Sir Isaac Newton (1642–1727). It takes a force of about 1 N to lift an apple.

FIGURE 7.1.1
A force is a push or a pull that acts between two objects.



What forces do

All around you, things move, or are in motion. When something changes the way it is moving, a force has acted. Some examples are shown in Figure 7.1.2.

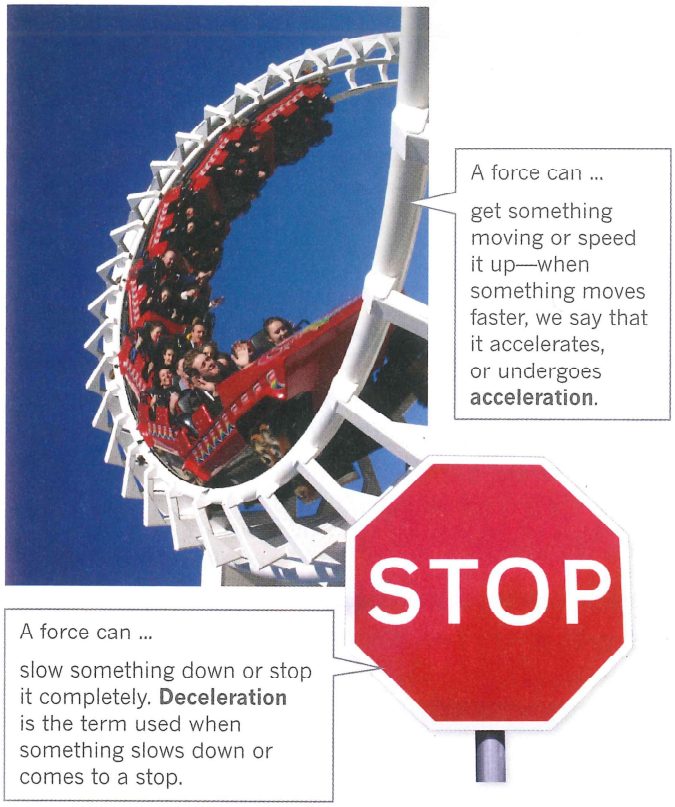


FIGURE 7.1.2 Forces are needed to cause a change in the motion of an object.

AB 7.6

Measuring forces

A spring can be used to measure a force. A spring is a spiral of wire that returns to its original size and shape after being stretched or squashed when the force is removed (Figure 7.1.3).

FIGURE 7.1.3 A force can stretch or squash a spring. When the force stops, the spring returns to its original size and shape.

The larger the pulling force, the more the spring is stretched and the higher the reading on the scale. The spring balance shown in Figure 7.1.4 operates in this way. Some bathroom and kitchen scales use a spring that is squashed or compressed to measure force. Digital scales rely upon objects squashing a special type of crystal called a piezoelectric crystal. This is a crystal that produces an electric current when squashed. The larger the current, the more the object weighs.

FIGURE 7.1.4 When you squash or stretch a spring or compress a special type of crystal, you can measure the size of a force.

Drawing forces

Many forces can act on an object at the same time. You can show these forces in a diagram such as Figure 7.1.5 by representing each as an arrow. The size of the arrow represents the size of the force, so the bigger the arrow, the bigger the force. Force A is the largest of these three forces and force C is the smallest.

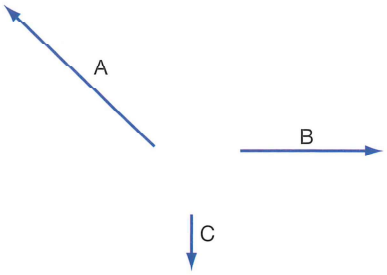


FIGURE 7.1.5 Forces are represented by arrows that show the direction in which the force acts.

Balanced forces

The forces acting on an object can be balanced or unbalanced. When sitting on a chair, the force of gravity pulls you downwards, towards Earth. The chair supports you by pushing upwards. This balances the downwards force of gravity. This balancing act is shown in Figure 7.1.6.

FIGURE 7.1.6 The downwards force of gravity is balanced by the upwards support force from the chair. The forces acting on this person are balanced.

Balanced forces don't always mean that the object is stopped—it might be travelling at the same speed without changing directions (Figure 7.1.7). Consider Nishika, about to ride her bike. To take off, Nishika must accelerate, pedalling quickly and hard to produce a force large enough to push her forwards. Another force is acting against her: friction. Forces of friction could slow her down, such as the roughness of the road and the air she pushes through. To speed up, Nishika keeps accelerating. Her pedalling needs to provide a driving force that is bigger than the force of friction acting in the opposite direction. As Nishika continues her journey, she may travel at a constant speed without slowing down or speeding up. When this happens, the driving force from her pedalling is cancelled out by the friction forces pushing her backwards. At this stage, her motion is constant and all of the forces acting on her are balanced.

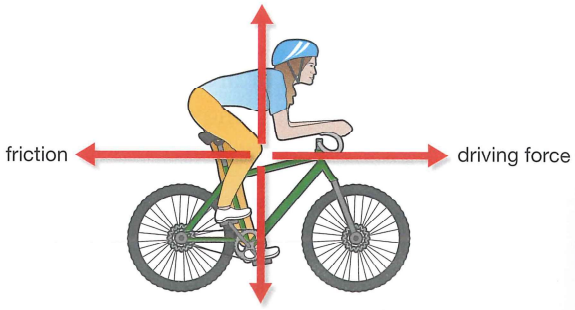


FIGURE 7.1.7 All the forces acting on Nishika are balanced: the driving force caused by her pedalling, the friction acting against her, the upwards support from the bike seat and the downward force of gravity.

Unbalanced forces

Whenever the forces acting on Nishika are unbalanced, her motion will change. Motion always changes in the direction of the unbalanced force, as can be seen in Figure 7.1.8. This means that the forces are unbalanced when Nishika:

- starts moving (by pedalling fast and hard)
- speeds up (by pedalling faster and harder)
- slows down (by using the brakes)
- comes to a stop (by using the brakes)
- changes direction (by turning the handlebars).

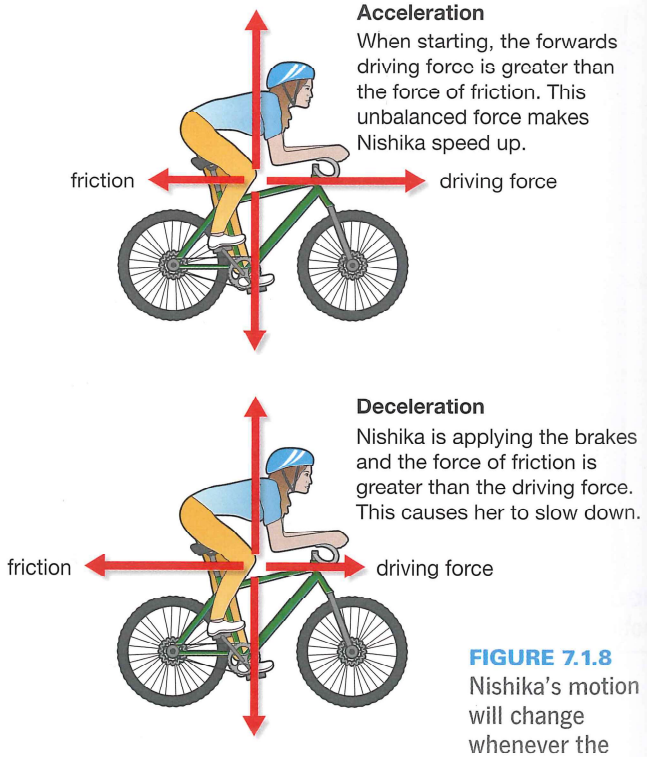


FIGURE 7.1.8 Nishika's motion will change whenever the forces acting are unbalanced.

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AB 7.2

Inertia

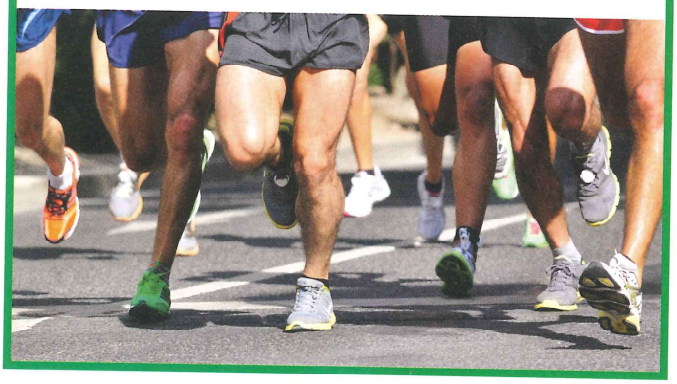
When you put your schoolbag down in your bedroom, it will stay there until something happens to it. Someone could lift it, push it or pull it to make it move; but if left alone, your schoolbag will stay as you left it. This ability of the schoolbag to remain unchanged is called its **inertia**. Everything and everyone possesses inertia. Inertia can be described as the tendency to resist any change in motion.

SciFile

Runaway success

In 1924, in Germany, Adolf (Adi) Dassler produced an athletic shoe made from goatskin, leather and chrome. Naming the shoe after himself, he called it Adidas. Jesse Owens wore Adidas shoes when he won four gold medals at the 1936 Olympic Games. This helped to make the brand famous.

Adi's older brother, Rudolph, broke away from the family company after a dispute. He established the Puma company in 1948.



Reducing impact force

Forces act on you all of the time. If these forces are balanced, then you will be still, or moving at a constant speed. If the forces are unbalanced, then your motion will change. Some of the forces that act on you in a collision can be large enough to break bones or damage internal organs.

Reducing impacts in sport

If you are riding a skateboard along the footpath and catch a wheel on a stone then your skateboard can stop moving. However, *you* will continue moving forwards until something makes you stop. Unfortunately this is likely to be the hard surface of the footpath ahead.

A properly fitted helmet, mouthguard and elbow, wrist and knee pads reduce the size of impact forces on you when something goes wrong while skateboarding, rollerblading or rollerskating. These accessories absorb some of the impact force in a collision and reduce the risk of serious injury (Figure 7.1.9).

FIGURE 7.1.9 Professional skateboard riders wear helmets and pads to lessen the impact force in a collision.

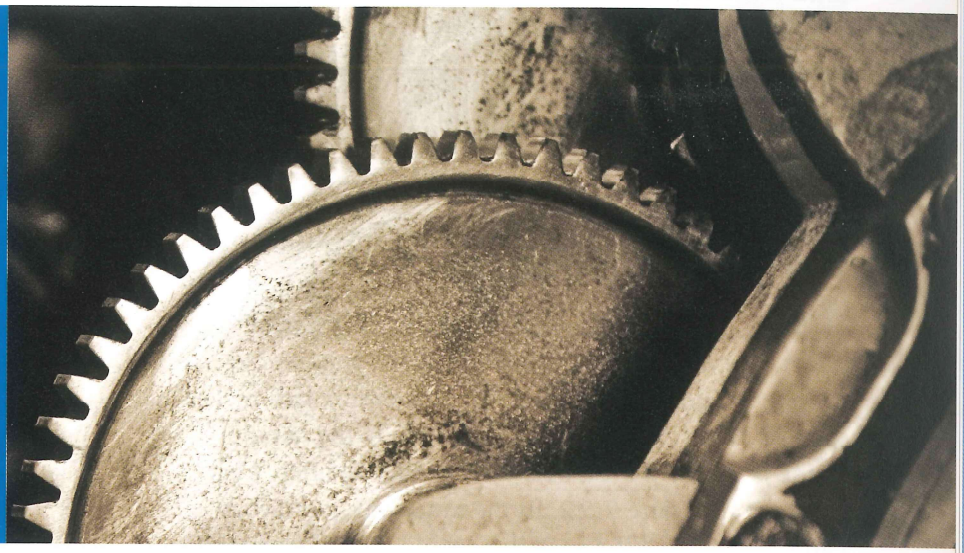
Footwear is also designed to reduce the force involved in the collision between a person's foot and the ground when walking, running or playing sport. The type of shoe worn needs to be suited to the type of activity. Running shoes are designed to cushion the toe and heel because it is these parts of the foot that are most affected by the impacts involved while running. Cushioned heels also spread the force over a larger area of the sole of the foot. Figure 7.1.10 shows the cushioning support of a running shoe.

AB 7.5

FIGURE 7.1.10 The cushioned sole of a shoe absorbs impact from running forces. A raised arch prevents the foot from rolling inwards.

MODULE
7.5 Simple machines

Whenever you use scissors or use a handle to open a door, you are using a simple machine. Simple machines include levers, ramps, wedges, screws, pulleys, wheels and gears. Simple machines make a job easier by changing the size, direction or speed of a force.



Machines and force

A simple machine is a device that makes a job easier to do. For example, the job of lifting the sound equipment in Figure 7.5.1 into the van is a difficult one—the equipment is heavy so a large force is needed to lift it. The job is much easier if a ramp is used—the force required is reduced and so the task becomes easier.

FIGURE 7.5.1 Loading and unloading heavy sound equipment would be almost impossible without the use of a ramp.

science 4 fun

Turning on a tap

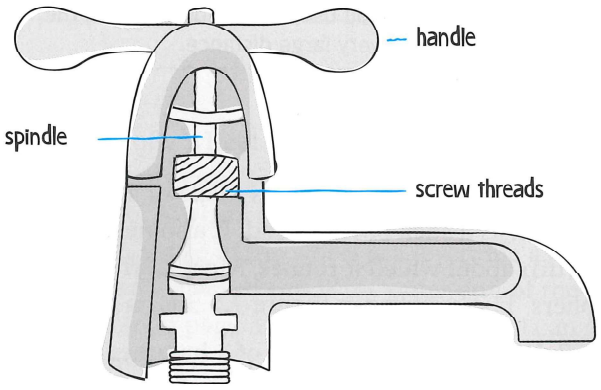
Can you turn on a tap without its handle?

Collect this ...

a tap that has a handle that can be unscrewed

Do this ...

1. Place a plug in the sink to avoid losing any parts down the drain.
2. Carefully unscrew a tap.
3. Place the screw, handle and cap on the basin.
4. The tap should now just be a spindle, or an axle.
5. Try to turn on the tap using just your hands.
6. Offer a challenge to your friends and family. Can it be done?



Record this ...

1. Describe what happened.
2. Explain why you think this happened.

The force you need to apply to get a job done is called the **effort**. In this case, a ramp reduces the effort needed to lift the sound equipment into the van. The **load** is the force actually required to do the job. For the sound equipment, the load is its weight.

Simple machines work in three different ways. They can:

- change the size of a force
- make things speed up
- change the direction of a force.

Force multipliers

Lifting a car to change a tyre would require a huge effort. For most of us, this task would be impossible despite it being a small distance. However, a jack makes the task easier. You apply a small force to the handle of the jack, but wind it over a very long distance. This is shown in Figure 7.5.2. In the same way, a crowbar can be used to shift a rock that would otherwise be too heavy to move. Machines used in this way magnify the force you apply to do a job. These machines are known as **force multipliers**.

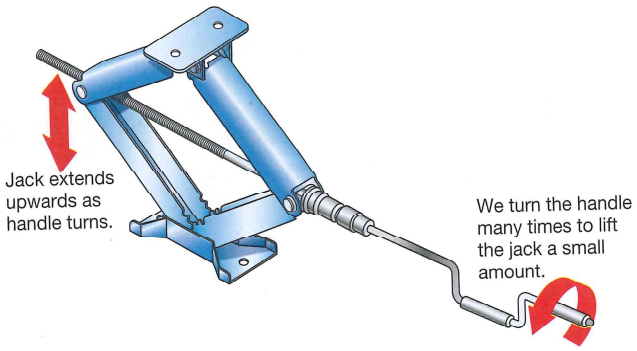


FIGURE 7.5.2 To lift a car you use a small force to turn the handle of a jack through a very large distance.

Levers

A **lever** is a simple machine that is made of a long, rigid object , such as a stick or metal rod. It also needs a pivot or **fulcrum** about which it rotates. Most levers are force multipliers. The crowbar in Figure 7.5.3 is an example.

Figure 7.5.4 shows how the claw of a hammer can be used as a lever to pull a nail (the load) out of a piece of wood. If you used just your fingers, then the effort required would be so high that the task is nearly impossible. Using a teaspoon to remove the lid of a coffee tin multiplies force in the same way (Figure 7.5.5).

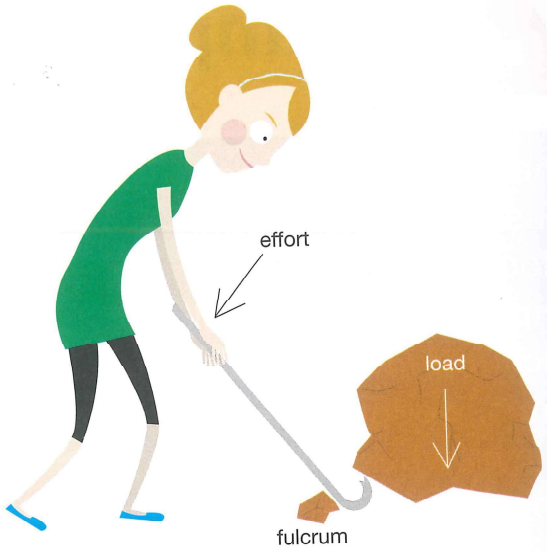


FIGURE 7.5.3 A crowbar acts as a lever. The lever increases the force that you can supply making it easier to move the rock.

FIGURE 7.5.4 Using a hammer as a lever reduces the effort needed to pull a nail from a piece of wood.

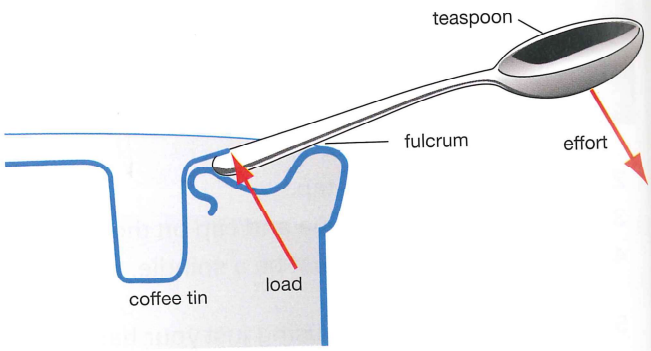


FIGURE 7.5.5 The teaspoon is a lever that multiplies the force, making it easier to remove the lid from the tin.

When you use a lever as a crowbar, or use a teaspoon to lift a lid from a tin, you are using a **first-class lever**. First-class levers have the fulcrum positioned between the effort and load forces. The trolley shown in Figure 7.5.6 is a first-class lever because the fulcrum (the axle that has the wheels attached) lies between the load that is being carried and the effort force the woman applies to the trolley. When you use a first-class lever, the lever moves through a larger distance than the load. In doing so, the force applied is increased. Other first-class levers are pliers, tin snips, hedgecutters and scissors.

If the load is positioned between the fulcrum and the effort, then the lever is called a **second-class lever**. This also acts as a force multiplier. A wheelbarrow is an example of such a lever. By lifting the handles a greater distance than the load, the force applied is increased. This is shown in Figure 7.5.7. Other second-class levers include bottle openers, paper guillotines and nutcrackers.

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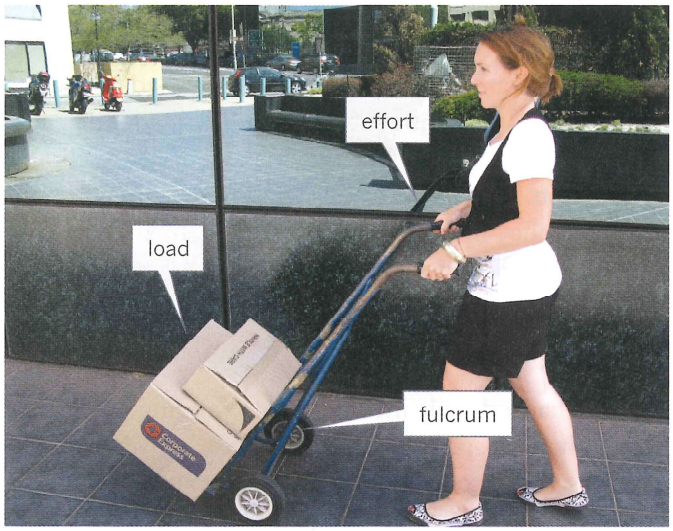
AB 7.9



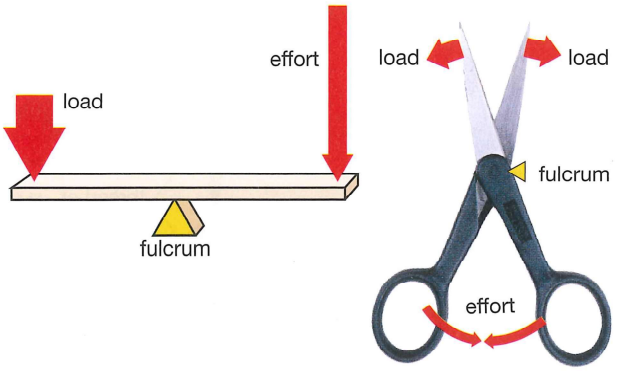


FIGURE 7.5.6 A first-class lever has its fulcrum positioned between the effort and the load. These levers act as force multipliers.

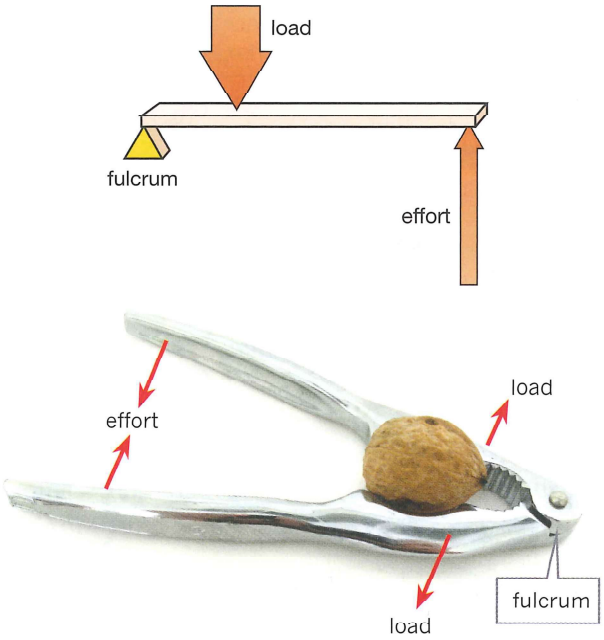


FIGURE 7.5.7 A second-class lever has its fulcrum positioned at one end of the system and the effort at the other. These levers also act as force multipliers.

Wheels

The centre of a **wheel** is called its **axle** and the outside of a wheel is its **rim**. This is shown in Figure 7.5.8 on page 326. A wheel on an axle is a special type of lever—each spoke of the wheel is a lever, the axle is the fulcrum and the wheel rim is the outer end of the lever. Figure 7.5.9 on page 326 shows how a wheel can act as a force-multiplier. When you apply a force to turn on a tap, the handle acts as a wheel. It increases the force you apply at its centre, allowing the axle to turn.

STEM p. 335

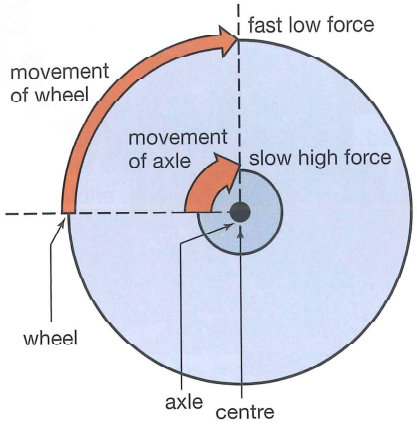


FIGURE 7.5.8 The rim of the wheel rotates with greater speed than the axle in its centre. As the wheel turns, the force at its centre is larger than the force at its rim.

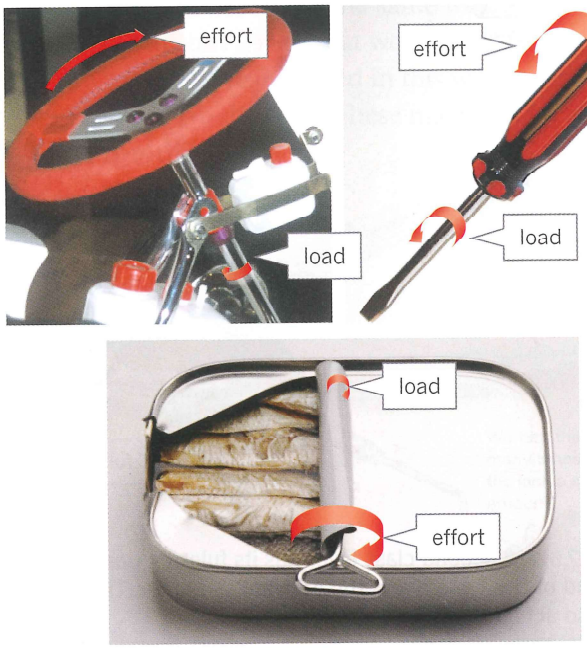


FIGURE 7.5.9 In each of these situations, a wheel is used to multiply force

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Gears

Gears are wheels that have teeth around their rim. Gears can mesh directly together, or be joined by a chain, such as on a bicycle or in the overhead cam shaft in some car engines. A set of gears that are connected is called a **gear train**. When one gear turns, the gear it interlocks with also turns, but in the opposite direction. The gear that supplies the force is called the **driving gear**. The gear that is connected to this gear is called the **driven gear**.

If the driving gear is smaller than the driven gear, then the combination acts as a force multiplier. The larger cog will rotate more slowly, but the force it delivers is much larger. This gear combination is called gearing down and is useful when you need a large force (Figure 7.5.10). For example, it would help a car get up a hill or help in winching it out when it is bogged.

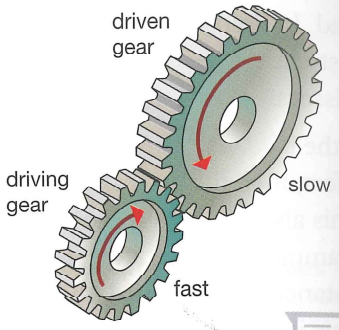

FIGURE 7.5.10 Gearing down makes the driven gear rotate more slowly than the driving gear, but with greater force.

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Inclined planes

While levers, wheels and gears can multiply a force to make it larger, **ramps** or **inclined planes** are force multipliers that *reduce* the effort force needed to do a job. Ramps make it easier to shift heavy sound equipment into vans. They also make it easier to climb a mountain. This is why roads wind up mountains. Ramps reduce the effort force required, by making you travel further. The longer the ramp, the less steep it will be. It will be easier to climb but you need to go further.

Screws

Prac 3 p. 333

A screw has a double inclined plane winding around it. This spiral is called the thread. A screw is easy to screw into wood but you need to turn it many times with your screwdriver. This can be seen in Figure 7.5.11.

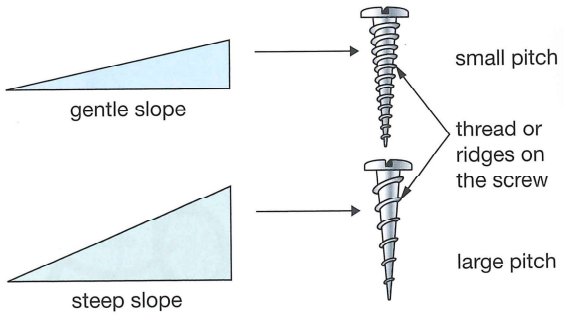


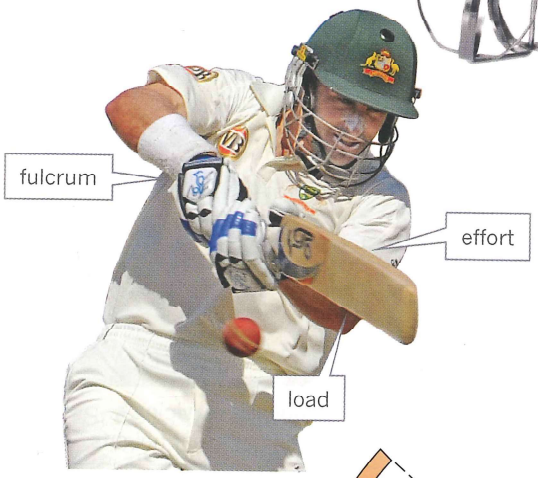
FIGURE 7.5.11 The pitch of a screw is the spacing between its thread. The screw with the small pitch has a longer spiral inclined plane and requires less force to be screwed into wood.

Speed multipliers

A machine can also make something move faster than otherwise possible. A machine such as the egg beater shown in Figure 7.5.12 is called a **speed multiplier**.

Tennis racquets, cricket bats and golf clubs are levers that are used as speed multipliers so that they hit balls at high speed. They are examples of **third-class levers**, with the effort positioned between the fulcrum and the load. This is shown in Figure 7.5.13.

FIGURE 7.5.12
The beaters spin much faster than the handle turns in this speed-multplier.



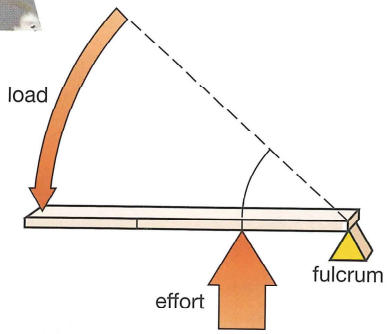


FIGURE 7.5.13
Cricket bats are third-class levers. They are speed multipliers.

AB 7.8

When using a broom, you apply a large effort force to the handle. The handle moves though a much shorter distance than the end of the broom, which moves much faster, but with less force. The broom acts as a third-class lever. It has traded an increase in force for an increase in speed.

Gears can also act as speed multipliers. If the driving gear is larger than the driven gear then the combination acts as a speed multiplier. This happens because one turn of the large cog makes the smaller cog spin a number of times. This gear combination is called gearing up. It is useful when using an eggbeater, where you require the beaters to spin faster than the handle turns. This is shown in Figure 7.5.14.

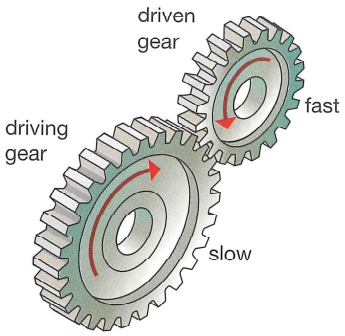


FIGURE 7.5.14
Gearing up makes the driven gear spin faster but with less force.

Changing the direction

A machine can also change the direction in which a force acts.

Wedges

Imagine trying to bite into an apple without your front teeth. Your front teeth, called *incisors*, cut like a knife. These are simple machines called **wedges**. A wedge is a double inclined plane that moves through another object. Wedges are used to split objects because they change the direction of a force by 90° and increase its size. When an axe is pushed downwards, the wedge of the axe head pushes the wood apart, as shown in Figure 7.5.15. Pins, needles, nails and doorstops are all examples of wedges.

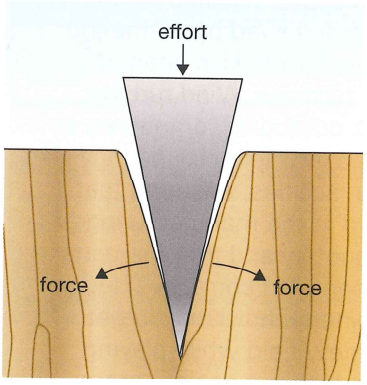


FIGURE 7.5.15
A wedge changes the direction of a force, splitting an object.

CHAPTER 7

Chapter review

Remembering

1 Which of the following statements are true and which are false?
- a Gravity is a contact force.
- b Weight is measured in kilograms.
- c The north pole of a magnet will attract the north pole of another magnet.
- d The magnetic field of a magnet is strongest at its poles.
- e A proton has a negative charge.
- f An electric field exists around a charged particle.

2 List four examples of a pushing force.

3 Name a surface that has little friction.

4 Is an electromagnet a temporary or a permanent magnet?

Understanding

5 Explain how a non-contact force can act on something without touching it.

6 Dropping a magnet could destroy its magnetism. Why?

7 If a comb pulled through your hair becomes negatively charged, predict the charge of your hair.

Applying

8 The diagrams shown in Figure 7.6.1 show forces acting on objects A, B, C and D.
- a Identify whether the forces are balanced or unbalanced in each case.
- b In the case of any unbalanced forces, predict the direction in which the object will move.

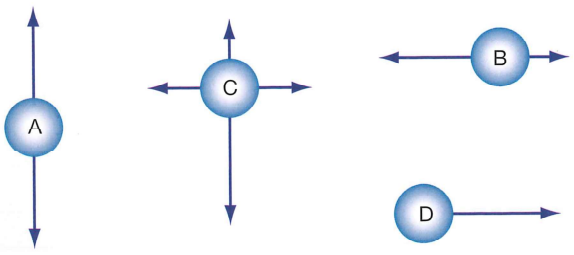


FIGURE 7.6.1

9 One gear is rotating anticlockwise. It is connected to a second gear. Predict the direction that the second gear will turn.

Analysing

10 Compare:
- a mass and weight
- b friction and gravity
- c the force between like poles of a magnet and the force between unlike poles
- d a force-multiplier and a speed-multiplier.

11 a State the class of lever acting in each situation in Figure 7.6.2. Letters represent the following: e = effort, f = fulcrum, l = load.
- b Classify which act as force multipliers and which act as speed multipliers.

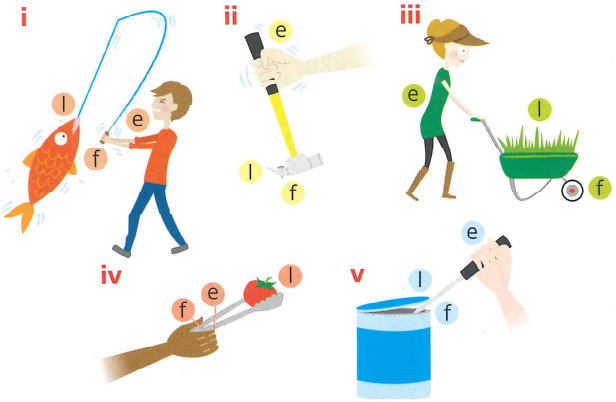


FIGURE 7.6.2

12 Analyse the combination of gears shown in Figure 7.6.3.
- a If gear 1 rotates in a clockwise direction, state which direction gears 2, 3 and 4 will rotate.
- b If axle 2 is rotated, identify whether gear 1 or gear 3 will turn faster.

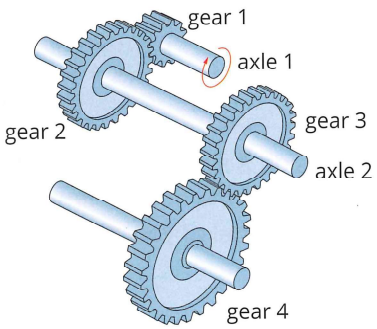


FIGURE 7.6.3